# Social Cybernetics for the Smart City: Collective Intelligence, Trust and Decisions in City Communities


Anton Kolonin

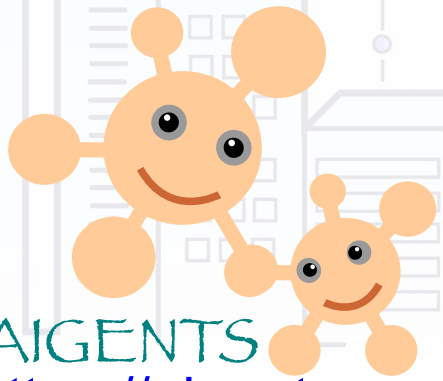
akolonin@aigents.com

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Novosibirsk State University

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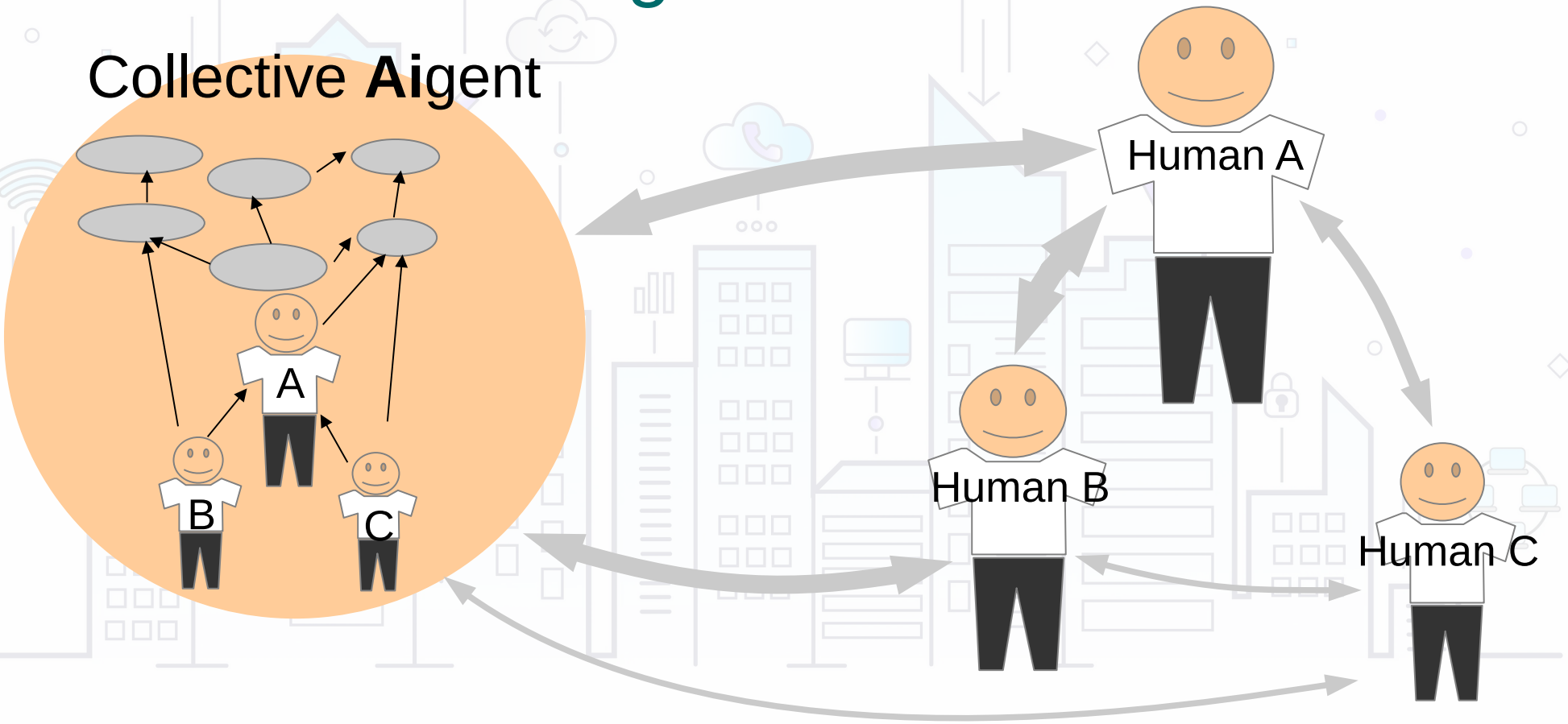



https://aigents.com

SingularityNET

https://singularitynet.io

# Decision support systems (DSS) relying on collective intelligence for communities




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# Computable cognitive-behavioral model based on social evidence and restricted by resources

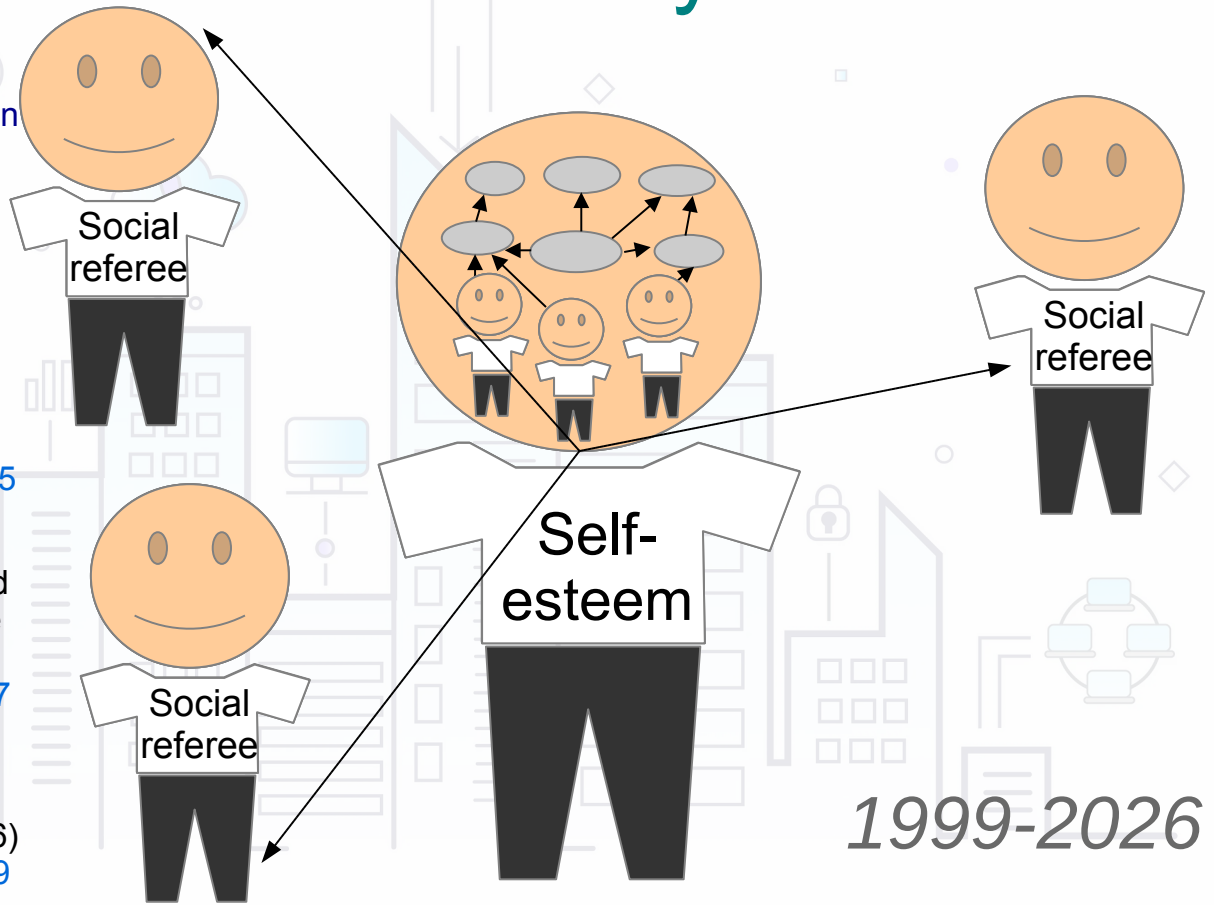
B.Goertzel, A.Kolonin, J.Pressing, C.Pennachin: Compassion-based artificial psyche design (1999-2000)
https://www.goertzel.org/benzine/WakingUpFromTheEconomyOfDreams.htm

A.Kolonin: Computable cognitive model based on social evidence and restricted by resources (2015)
https://ieeexplore.ieee.org/document/7361869

Anton Kolonin, Evgenii Vityaev, Yuriy Orlov: Cognitive Architecture of Collective Intelligence Based on Social Evidence (2016)
https://www.sciencedirect.com/science/article/pii/S1877050916317239

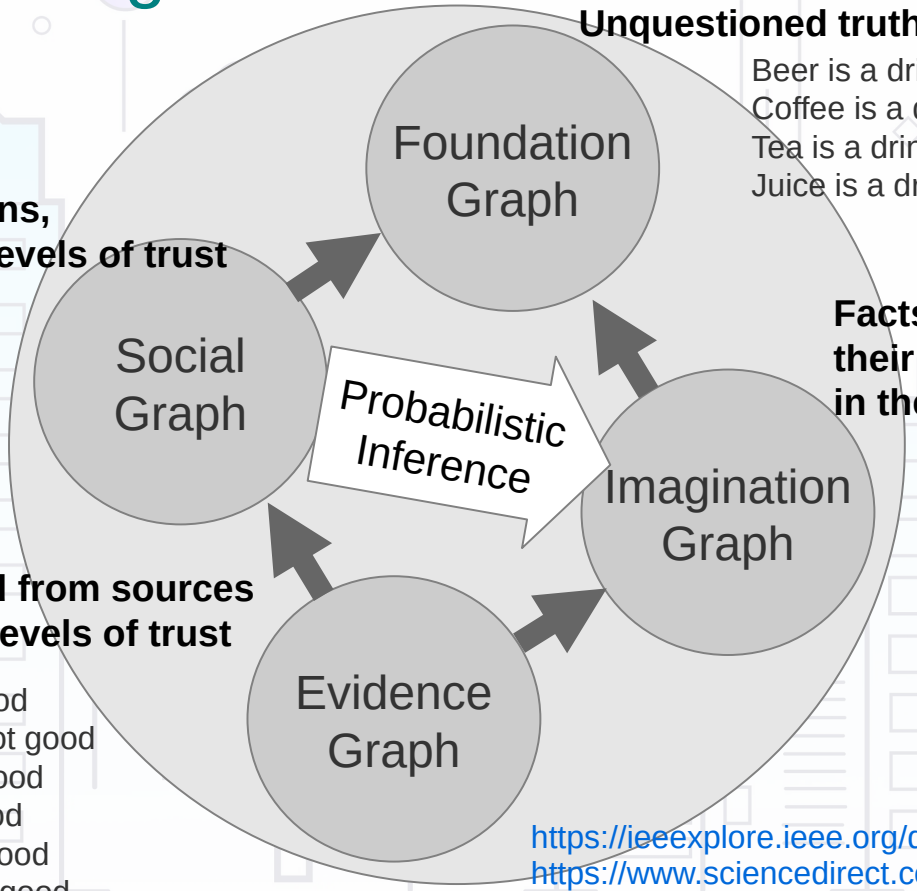
A.Kolonin: Resource-Constrained Social Evidence Based Cognitive Model for Empathy-Driven Artificial Intelligence (2018)
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

A.Kolonin, Cognitive architecture and behavioral model based on social evidence and resource constraints (2026)
https://link.springer.com/article/10.1186/s40708-026-00294-1




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# Social evidence-based resource-constrained cognitive-behavioral model




**Unquestioned truths and beliefs**

Beer is a drink
Coffee is a drink
Tea is a drink
Juice is a drink

**Social connections, with account to levels of trust**

Mary is my best friend
Peter is my friend
Bob is Peter's friend

**Facts weighed based on their social sources and assessments, in the light of unquestioned beliefs**

★★★ I like juice very much
★★ I like tea
★ I may try coffee
I won't try beer

**Facts obtained from sources with different levels of trust**

Bob said beer is good
Mary said beer is not good
Mary said juice is good
Mary said tea is good
Peter said juice is good
Peter said coffee is good

https://ieeexplore.ieee.org/document/7361869
https://www.sciencedirect.com/science/article/pii/S1877050916317239
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10
https://link.springer.com/article/10.1186/s40708-026-00294-1

*2018-2026*

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# Perception & Action driven by Believed Social Evidence

$i$ – person of consideration
$j$ – concept (action) of consideration (exposed evidence)
$l$ – person's belief item (of foundation graph of size $L$ - personal preference base)
$k$ – person's correspondent (of social graph of size $K$ - social reference base)
$B_{il}$ – person's $i$ mental attachment to $l$ (personal preference)
$S_{ik}$ – person's $i$ social bind to $k$ (social reference)

recursion

$R^B$ $R^S$

$$P_{ij} = \Sigma_{l=1,L}( E^B_{ijl} * B_{il} ) * \Sigma_{k=1,K}( E^S_{ijk} * S_{ik} )$$ *(what we think)*

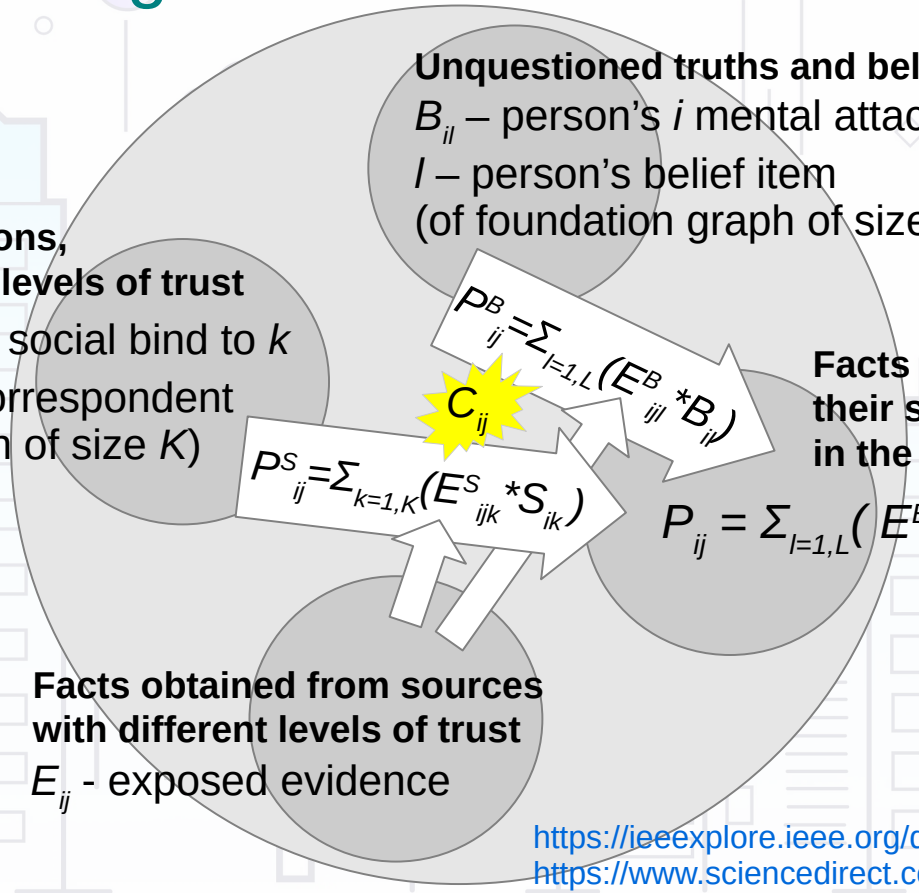
$E^B_{ik}$ – concept (action) $j$ agreement or compatibility with $l$ in mind of $i$ (believed evidence)
$E^S_{ik}$ – concept (action) $j$ expression or confirmation by $k$ in view of $i$ (social evidence)
$P_{ij}$ – concept (action) $j$ power for $i$ (personal evidence)

$$C_{ij} = L + K + R^B + R^S$$ *(why we don't think about that)*

$C_{ij}$ – concept (action) cost $j$ cost for $i$ (personal cost)

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# Social evidence-based resource-constrained cognitive-behavioral model




**Unquestioned truths and beliefs**

$B_{il}$ – person's $i$ mental attachment to $l$

$l$ – person's belief item
(of foundation graph of size $L$)

**Social connections,
with account to levels of trust**

$S_{ik}$ – person's $i$ social bind to $k$

$k$ – person's correspondent
(of social graph of size $K$)

**Facts weighed based on
their social sources and assessments,
in the light of unquestioned beliefs**

$$P_{ij} = \Sigma_{l=1,L}( E^B_{ijl} * B_{il} ) * \Sigma_{k=1,K}( E^S_{ijk} * S_{ik} )$$

**Facts obtained from sources
with different levels of trust**

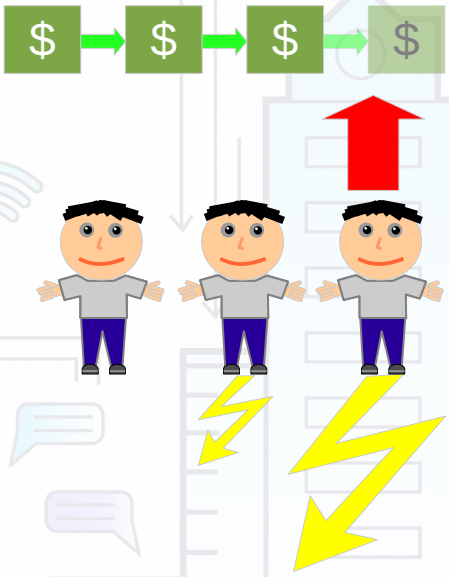
$E_{ij}$ - exposed evidence

https://ieeexplore.ieee.org/document/7361869
https://www.sciencedirect.com/science/article/pii/S1877050916317239
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10
https://link.springer.com/article/10.1186/s40708-026-00294-1

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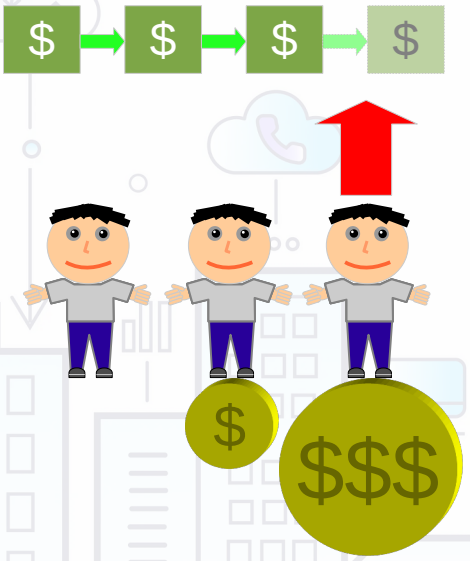
# Consensus – technology to govern distributed multi-agent systems such as blockchains or societies, resistant to takeover and scam.

## Proof-Of-Work



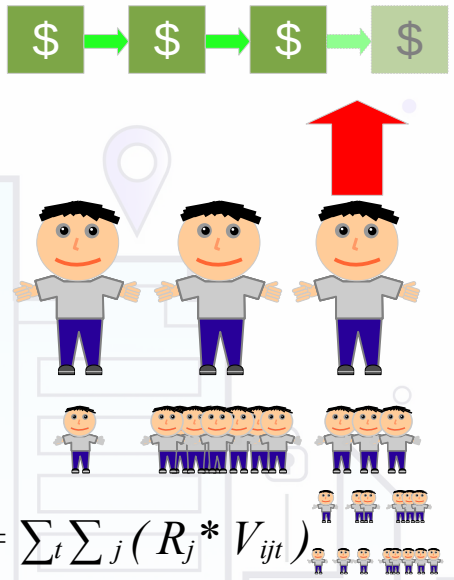

**Force is Power:**
Those who own more computing resources govern the network.

## Proof-Of-Stake




**Money is Power:**
Those who have more money govern the network.

## Proof-Of-Reputation




$$R_i = \sum_t \sum_j ( R_j * V_{ijt} )$$

**Reputation is Power:**
Those who earn a better reputation and a greater long-term audience base govern the network.

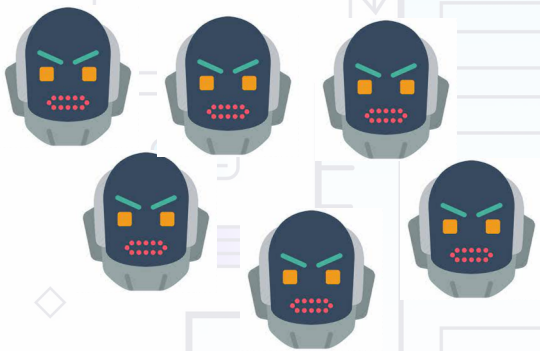
https://steemit.com/blockchain/@aigents/proof-of-reputation-as-liquid-democracy-for-blockchain

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# Real Pain: Resisting Reputation Gaming

Cheaters





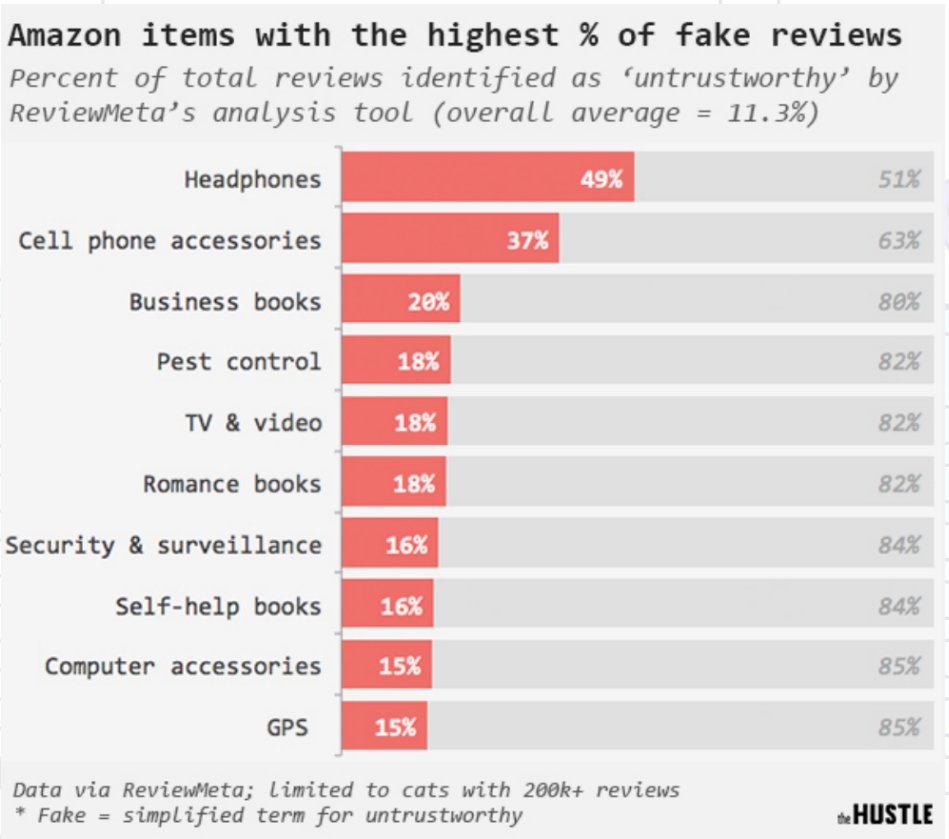
Fake accounts with churning and reputation gaming bots



Honest vendor



Honest buyers




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# Bad reputation (rating) system

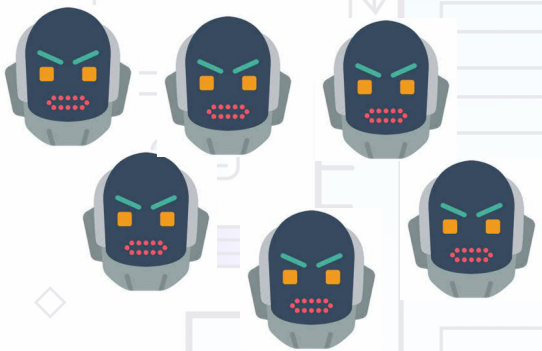


Cheaters earn





Honest vendors lose profits



Fake account and bot farm owners are sponsors are paid





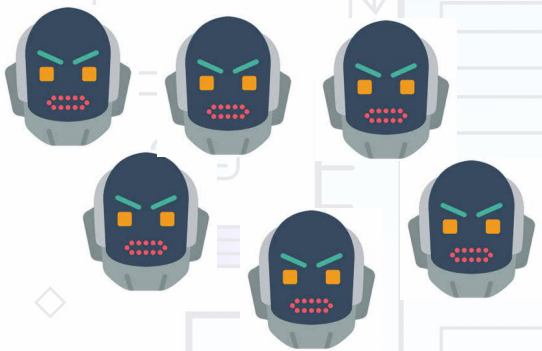
Honest buyers lose money

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# Right reputation (rating) system



Cheaters lose profits



Fake account and bot farm owners are sponsors are still paid for no purpose









Honest vendors gain profits



Honest buyers get goods

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# Marketplace imitation modeling

Cheaters



Fake goods

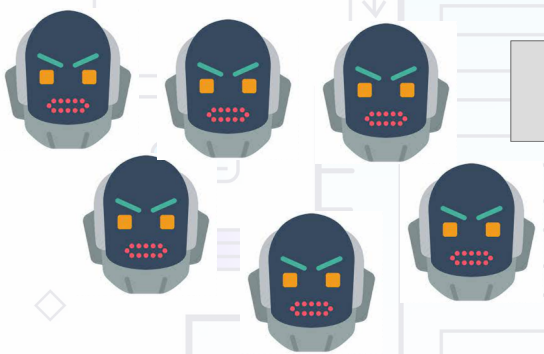
Virtual marketplace



Real goods



Honest vendor



Fake reviews

Real reviews



Fake buys

Real buys

Fake accounts with churning and reputation gaming bots

We select the parameters for the rating system based on a “weighted liquid rank” to maximize losses for cheaters and gains for honest vendors and minimize losses for honest buyers.

Honest buyers

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# Weighted Liquid Rank

https://arxiv.org/pdf/1806.07342

**Algorithm 1** Weighted Liquid Rank (simplified version)

**Inputs**:
1) Volume of rated transactions each with financial value of the purchased product or service and rating value evaluating quality of the product/service, covering specified period of time;
2) Reputation ranks for every participant at the end of the previous time period.

**Parameters**: List of parmeters, affecting computations - default value, logarithmic ratings, conservatism, decayed value, etc.

**Outputs**: Reputation ranks for every participant at the end of the previous time period.

1: **foreach of** *transactions* **do**
2: **let** *rater_value* be rank of the rater at the end of previous period of default value
3: **let** *rating_value* be rating supplied by trasaction rater (consumer) to ratee (supplier)
4: **let** *rating_weight* be financial value of the transaction of its logarithm, if logarithmic ratings parameter is set to true
5: **sum** *rater_value*rating_value*rating_weight* for every ratee
6: **end foreach**
7: **do** normalization of the sum of the muliplications per ratee to range *0.0-1.0*, get *differential_ranks*
8: **do** blending of the old_ranks known at the end of previous peiod with differential_ranks based on parameter of conservatism, so that *new_ranks* = *(old_ranks*conservatism+N*(1-differential_ranks))*, using decayed value if no rating are given to ratee during the period
9: **do** normalization of *new_ranks* to range *0.0-1.0*
10: **return** *new_ranks*

- $R_d$ - default initial reputation rank;
- $R_c$ - decayed reputation in range to be approached by inactive agents eventually;
- *C* - conservatism as a blending "alpha" factor between the previous reputation rank recorded at the beginning of the observed period and the differential one obtained during the observation period;
- *FullNorm* – when this boolean option is set to *True* the reputation system performs a full-scale normalization of incremental ratings;
- *LogRatings* - when this boolean option is set to *True* the reputation system applies *log10(1+value)* to financial values used for weighting explicit ratings;
- *Aggregation* - when this boolean option is set to *True* the reputation system aggregates all explicit ratings between each unique combination of two agents with computes a weighted average of ratings across the observation period;
- *Downrating* - when this boolean option is set to *True* the reputation system translates original explicit rating values in range *0.0-0.25* to negative values in range *-1.0* to *0.0* and original values in range *0.25-1.0* to the interval *0.0-1.0*.
- *UpdatePeriod* – the number of days to update reputation state, considered as observation period for computing incremental reputations.

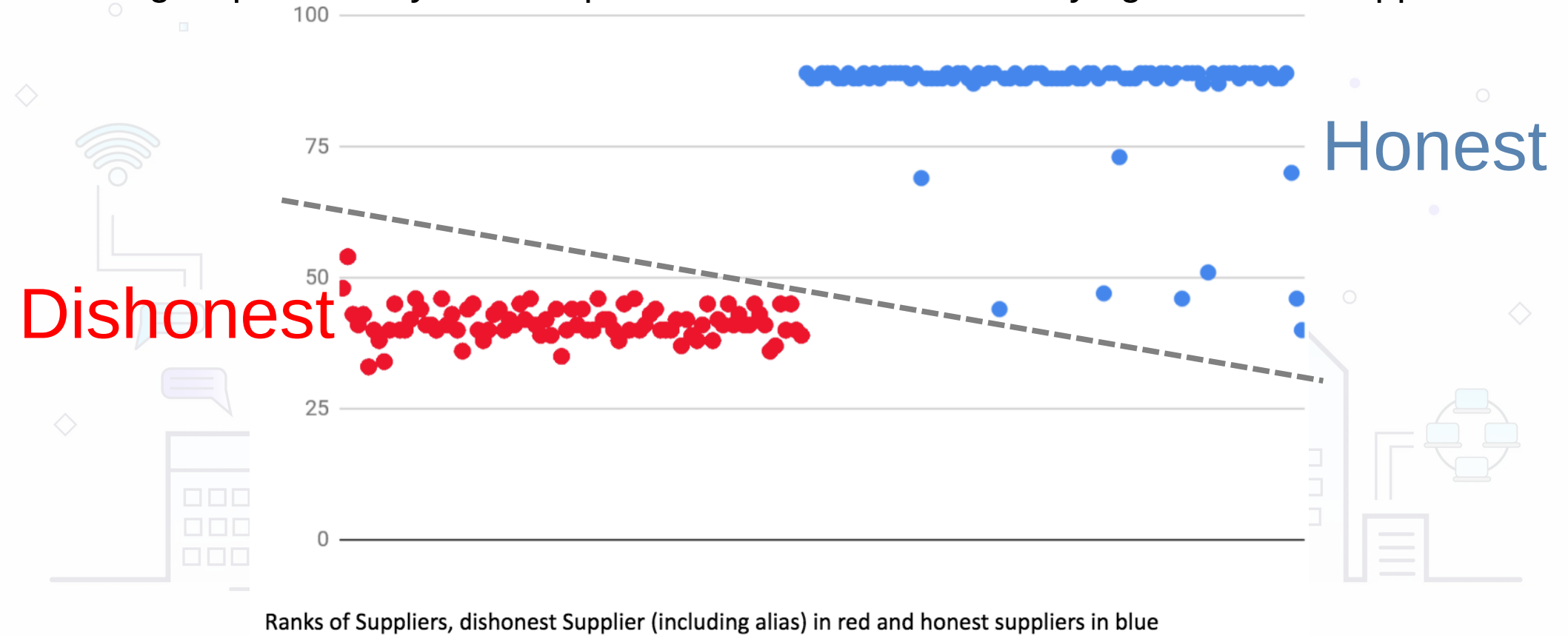
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# Reputation System for Marketplaces

https://medium.com/singularitynet/minimizing-recommendation-fraud-7dabbee8fc00

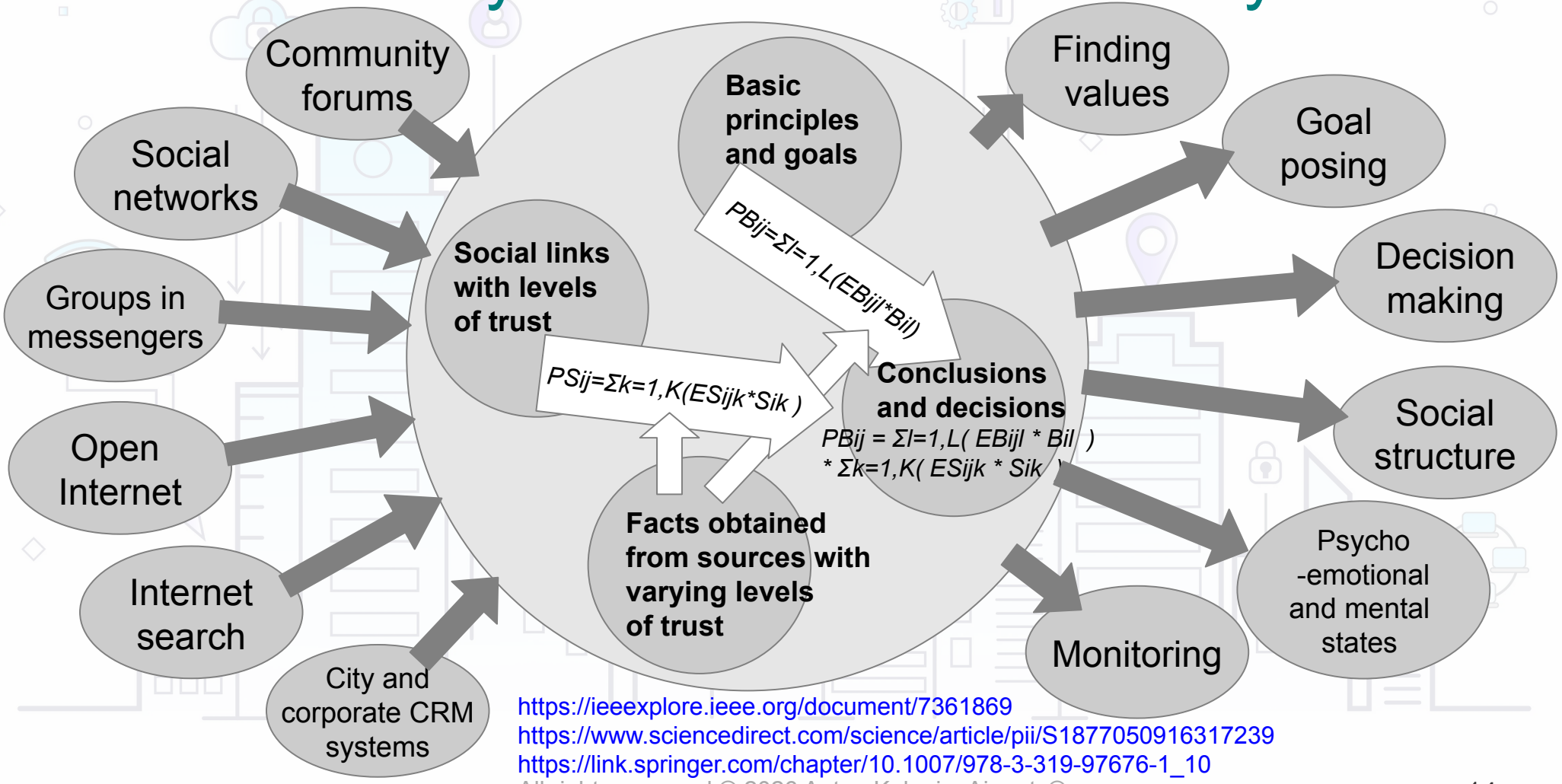
Using Reputation System for protection from scam identifying dishonest suppliers.




Ranks of Suppliers, dishonest Supplier (including alias) in red and honest suppliers in blue

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# Social Cybernetics for «Smart City»




https://ieeexplore.ieee.org/document/7361869
https://www.sciencedirect.com/science/article/pii/S1877050916317239
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

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# Aigents® - Social computing platform




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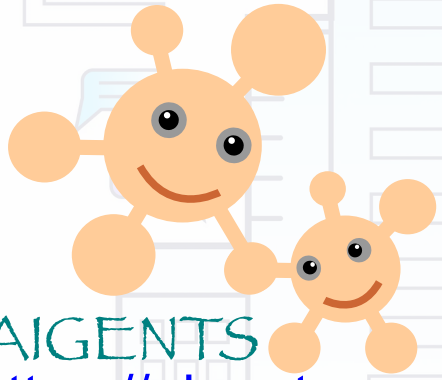
# Thank you for attention!

## More on YouTube:


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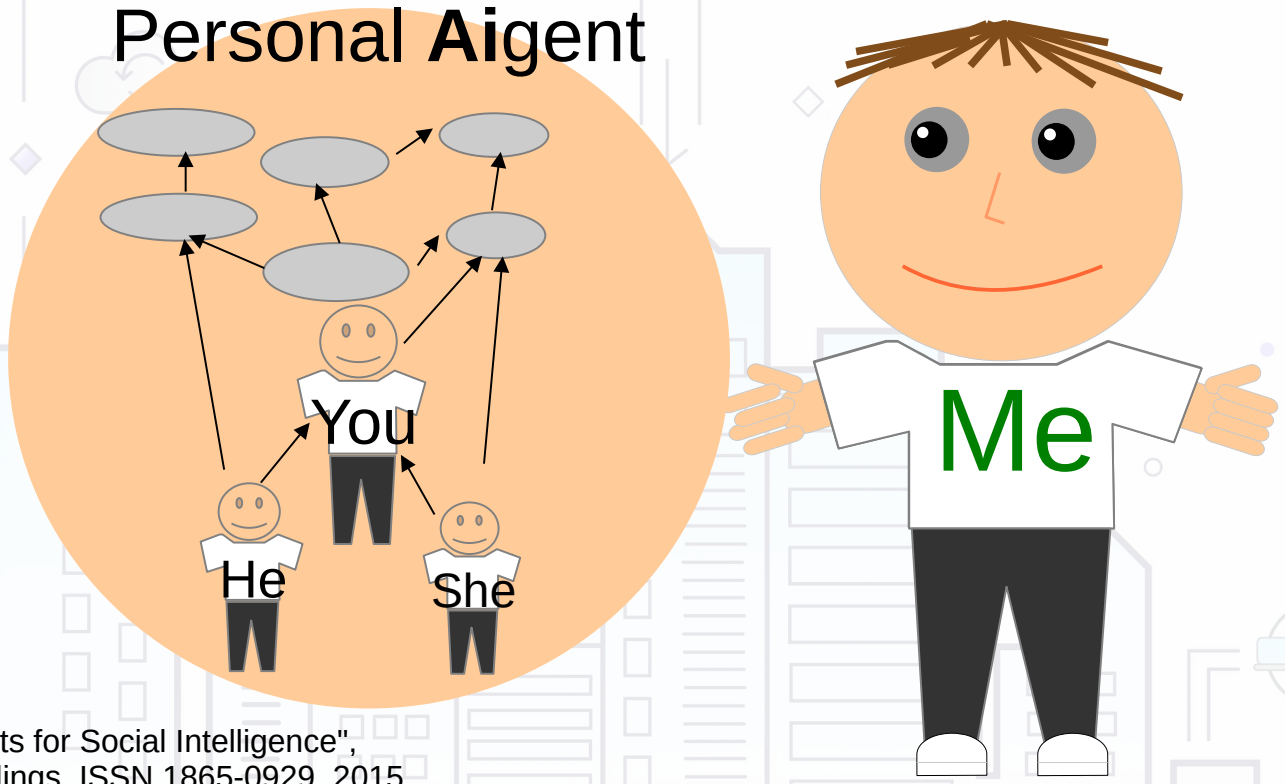


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# Aigents® - Personal Assistant



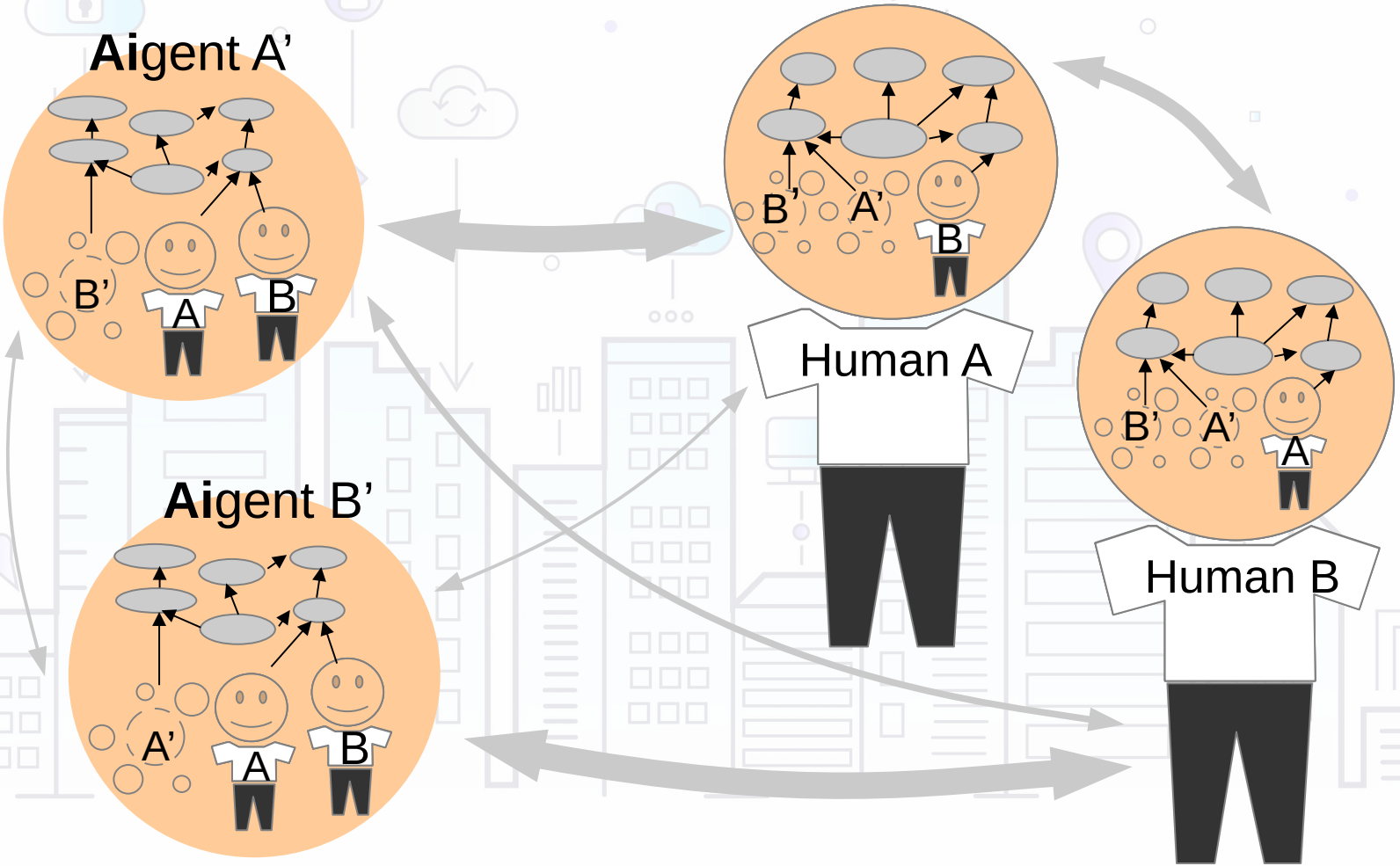

A. Kolonin, "Aigents: Adaptive Personal Agents for Social Intelligence", KESW, 6th International Conference Proceedings, ISSN 1865-0929, 2015
https://link.springer.com/chapter/10.1007/978-3-319-24543-0_22

A.Kolonin, "Adaptive experiential learning for business intelligence agents", CSGB Symposium Proceedings, 2016
https://ieeexplore.ieee.org/document/7587679

*2015-2016*

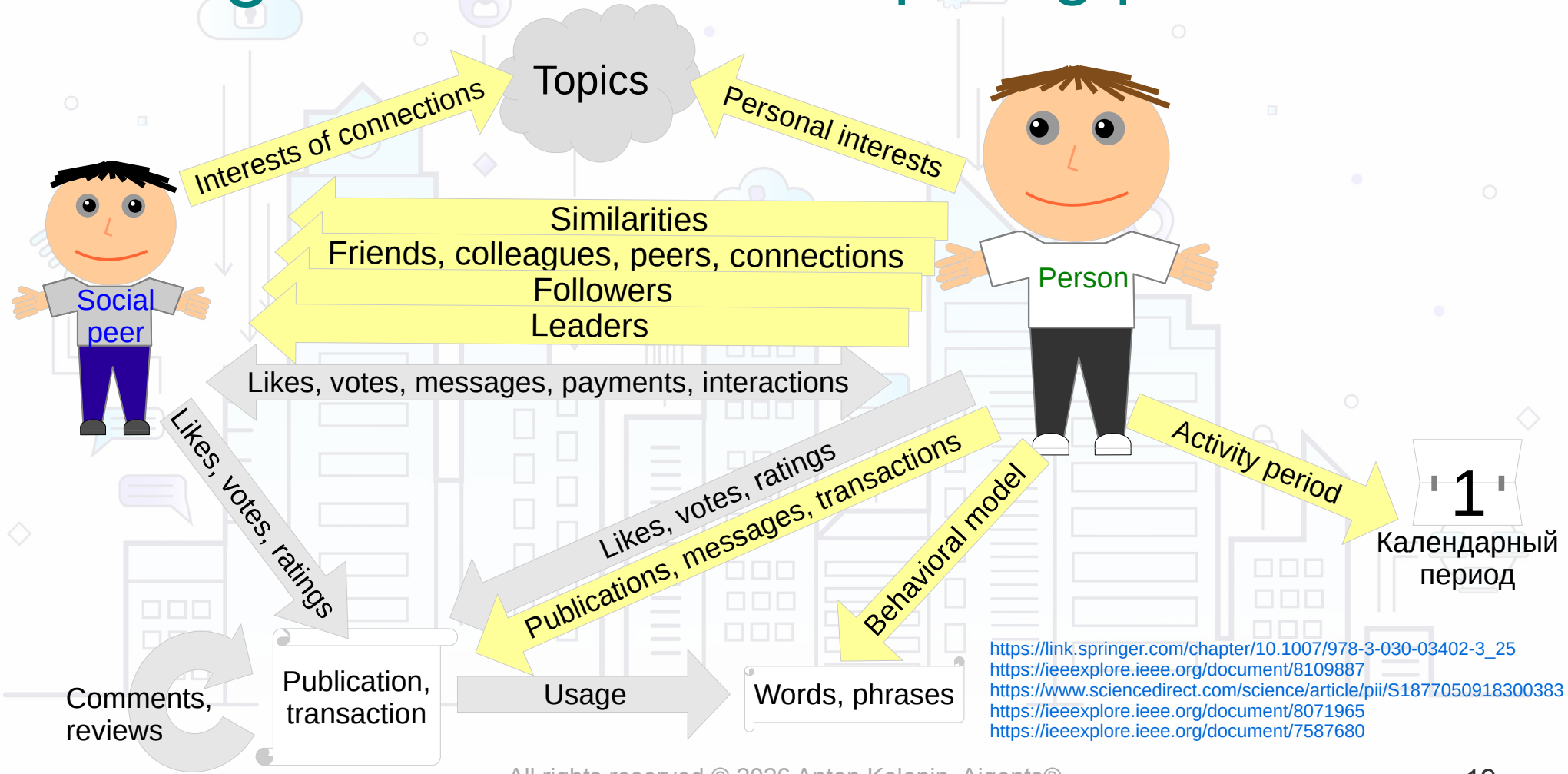
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# Socially-”aligned” emphatic AI-agents




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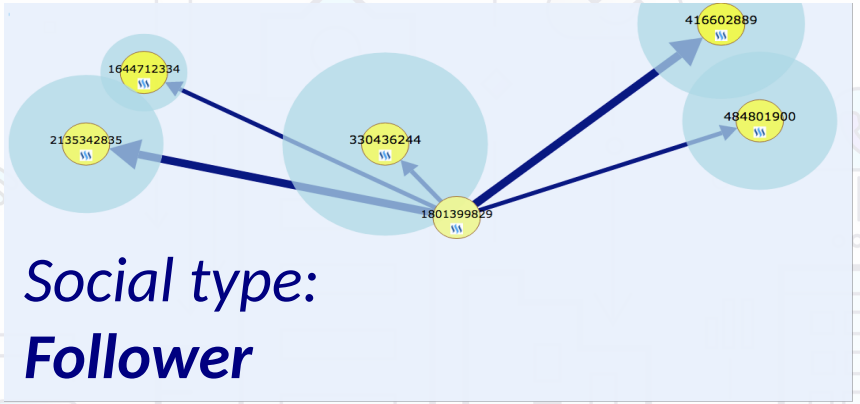
# Aigents® - Social computing platform




https://link.springer.com/chapter/10.1007/978-3-030-03402-3_25
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https://ieeexplore.ieee.org/document/7587680

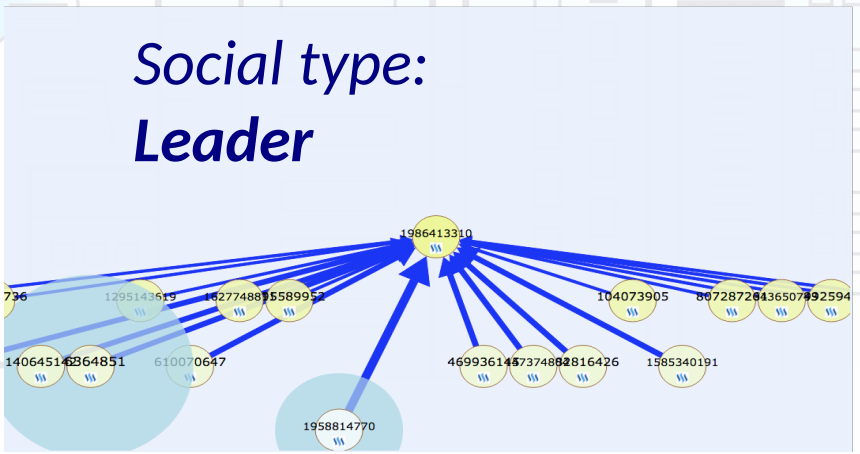
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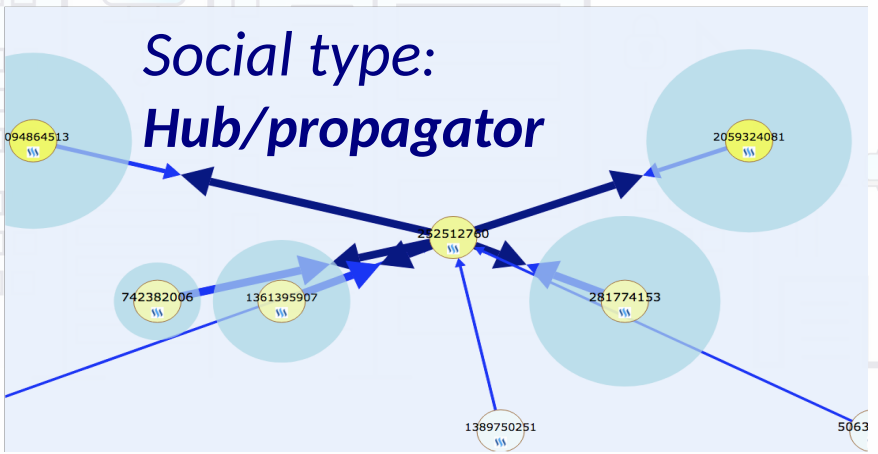
# Solution: Identification and discovery of socio-communicative types, profiles and structures, social circles and audiences, opinion leaders, and information dissemination channels.





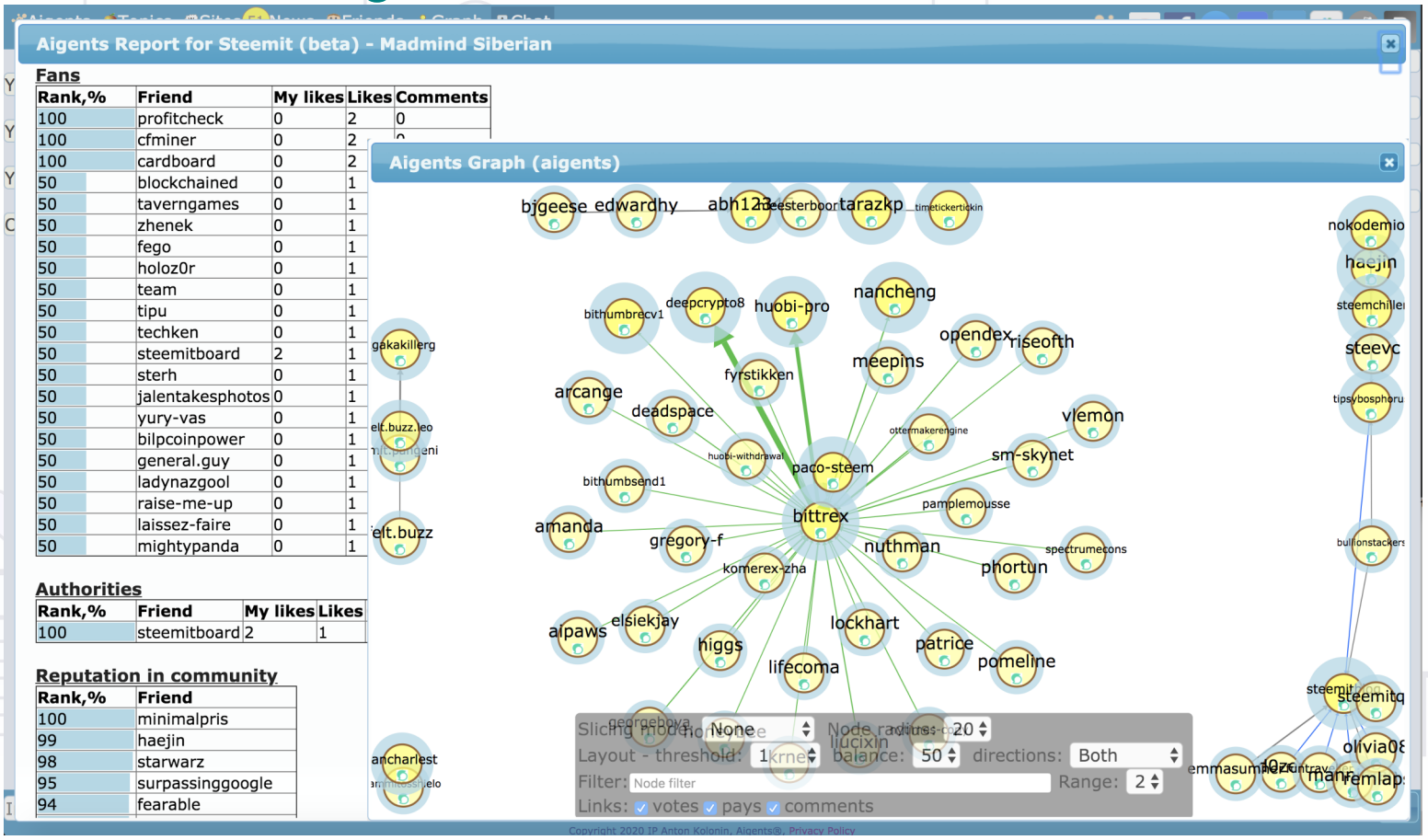





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# Solution: Quantitative assessment of reputation and trust levels, as well as the nature, quantity, and quality of interpersonal relationships within large-scale online communities.




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# Solution: News filtering, recommendation, and tagging based on sentiment, cognitive distortions and social relevance

News with positive (green) and negative (red) sentiment and social relevance (yellow)







Topics by news with positive (green) and negative (red) sentiment

https://blog.singularitynet.io/aigents-sentiment-detection-personal-and-social-relevant-news-be989d73b381

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# Automated detection of cognitive distortions, emotions and social situations

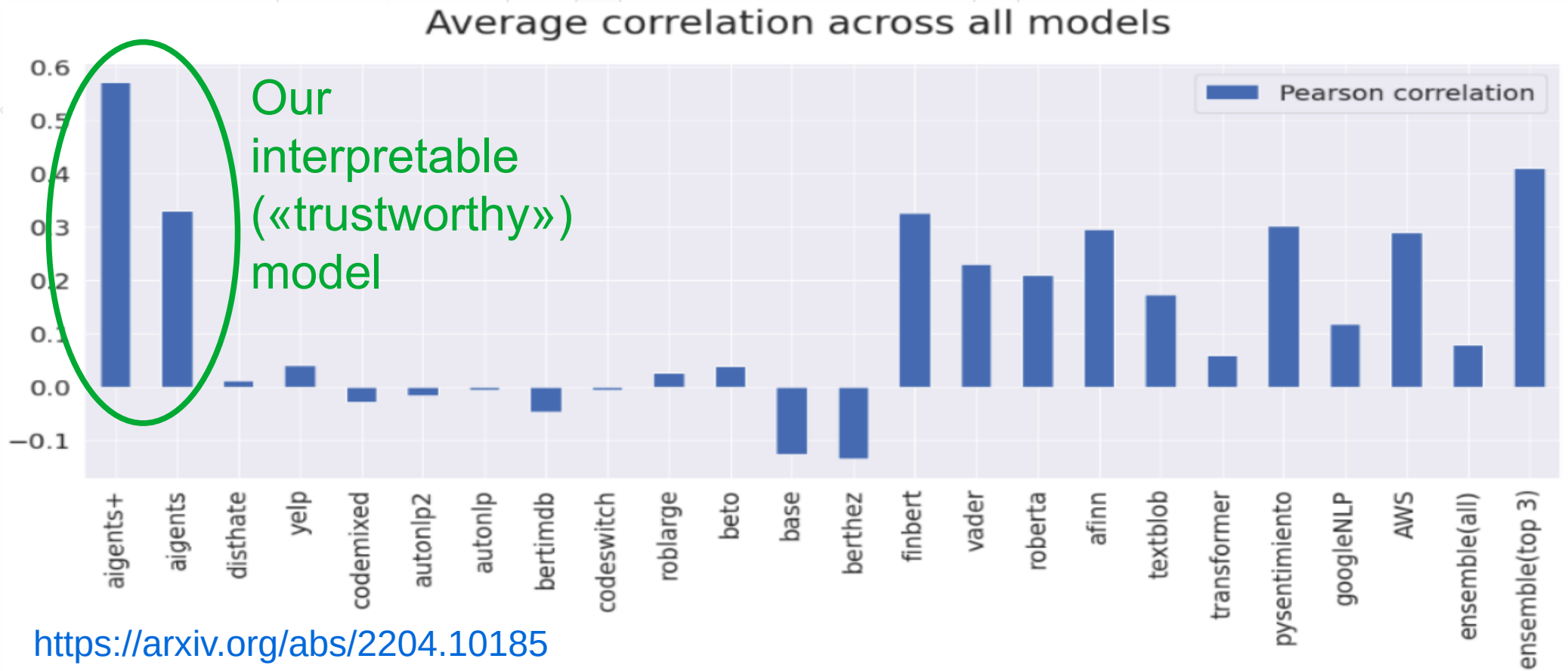
P: Hi! How are you?
K: Bad. They have kicked out the loser! (tentatively diagnosing #being fired and #personalization)
P: How did that happen? Who did that? (clarifying **social situation**)
K: It was my job. I am fired so to speak. (confirming #being fired)
P: Does it make you anxious or make you angry on them? (clarifying **emotion** - #anxiety or #anger)
K: Anxious, of course. I have to pay the mortgage. (confirming #anxiety, diagnosing #shouldment)
P: You have been fired from your job and you are afraid, that won't be able to pay, right? (#being fired, #anxiety)
K: Yes. (confirming diagnosis in respect to **social situation** and **emotion**)
P: What do you think was the reason?
K: I just was not lucky – the investor has left the company. (diagnosing #personalization)
P: Was that your fault? (clarifying **cognitive distortion** - #personalization)
K: Not at all. I just feel like my life is ruined. (change diagnosis to #catastrophizing)
P: You are anxious, that after the being fired you won't be able to find a new job, correct? (#anxiety, #being fired, #catastrophizing)
K: Exactly. Although in fact I am an experienced professional. (confirming diagnosis in regard to **emotion**, **social situation** and **cognitive distortion** moving on to correction with healing interventions)

https://doi.org/10.1016/j.procs.2022.11.054

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# How do you determine the tone or sentiment of a text?

Metrics: Sentiment, Positiveness, Negativeness, Contradictiveness

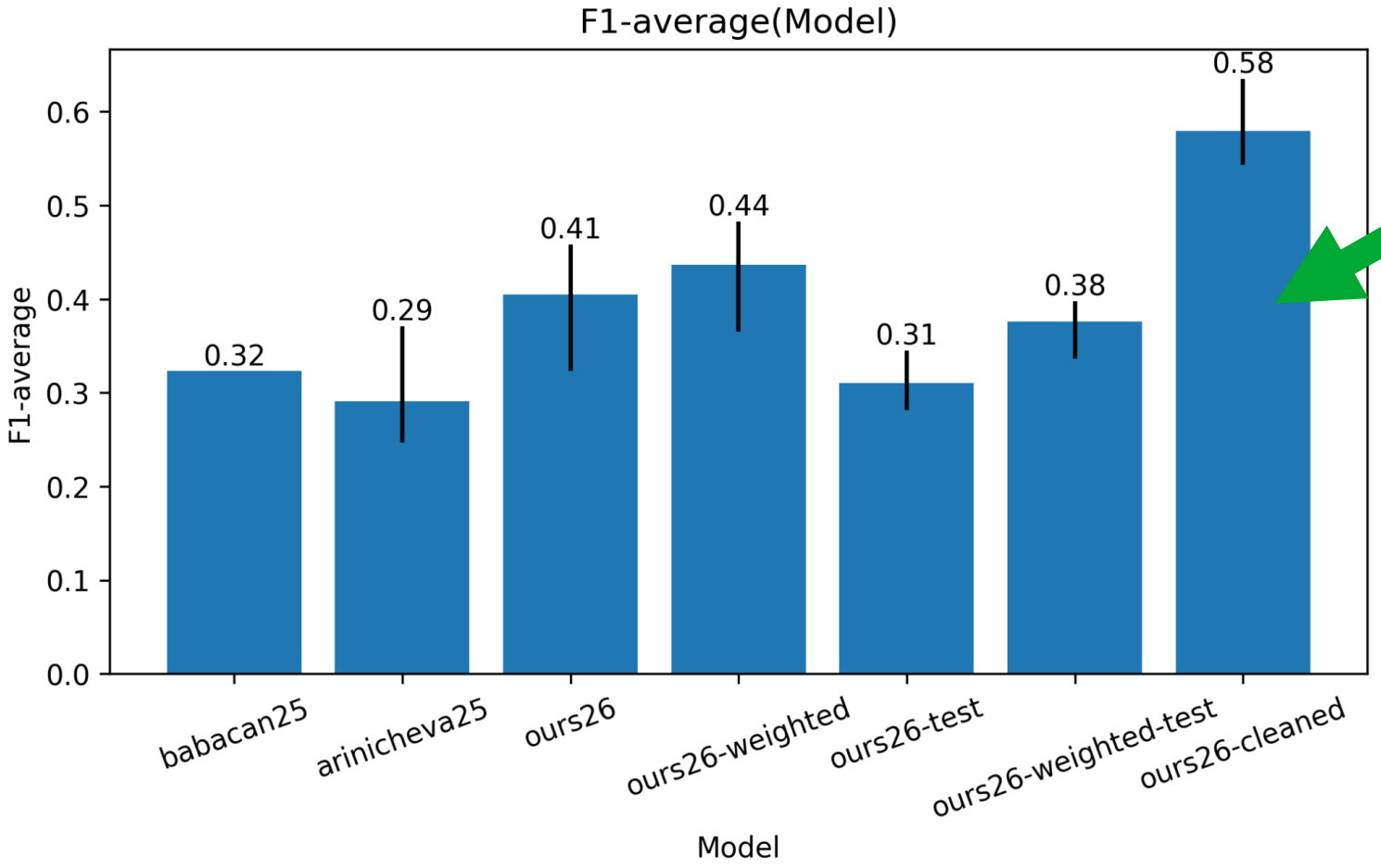



Our interpretable («trustworthy») model

https://arxiv.org/abs/2204.10185

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# Automated detection of cognitive distortions - state of the art

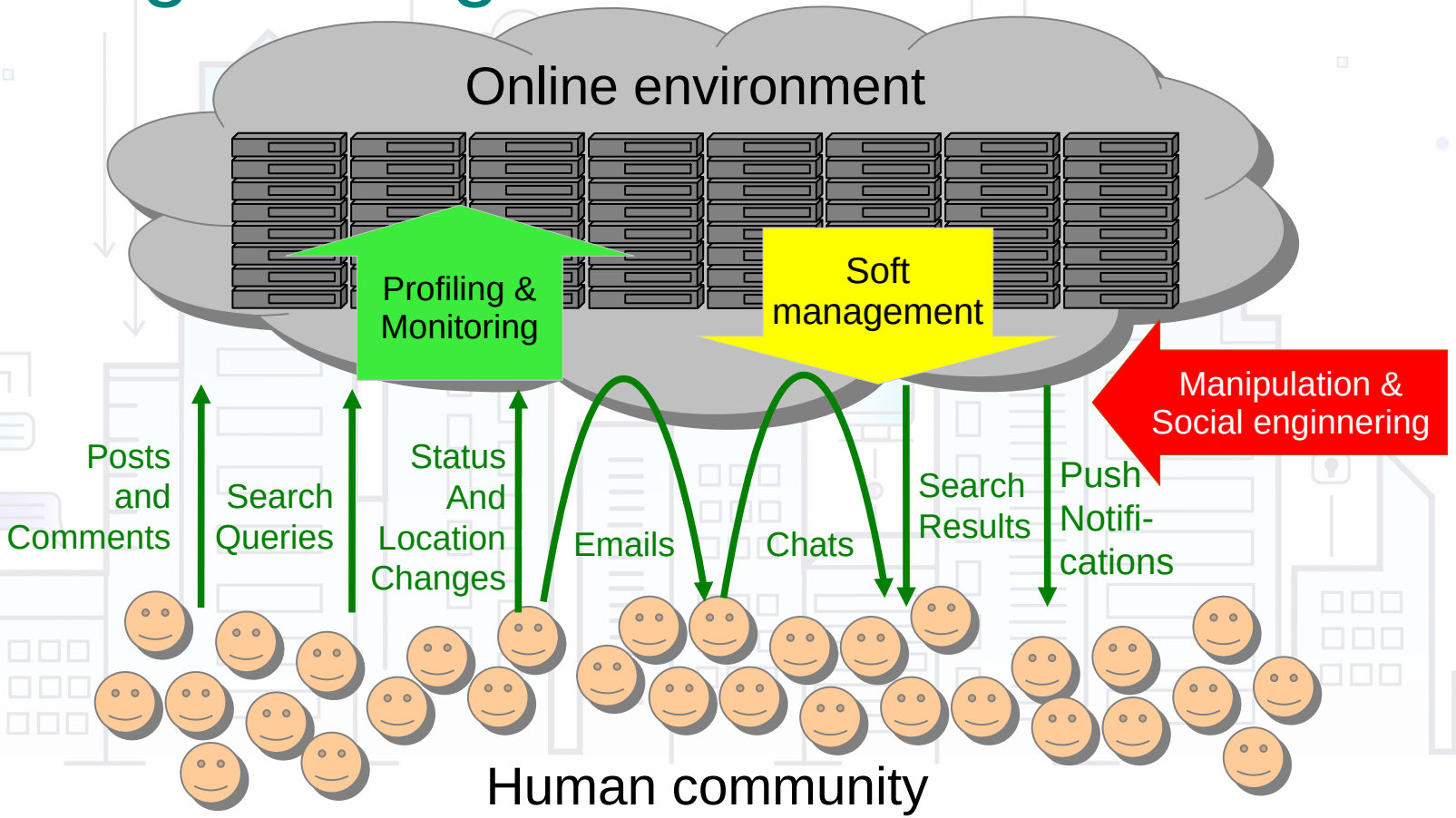



Our interpretable («trustworthy») model

https://peerj.com/articles/cs-3699/ https://arxiv.org/abs/2511.05969

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# Architecture of collective intelligence and social engineering in online environments

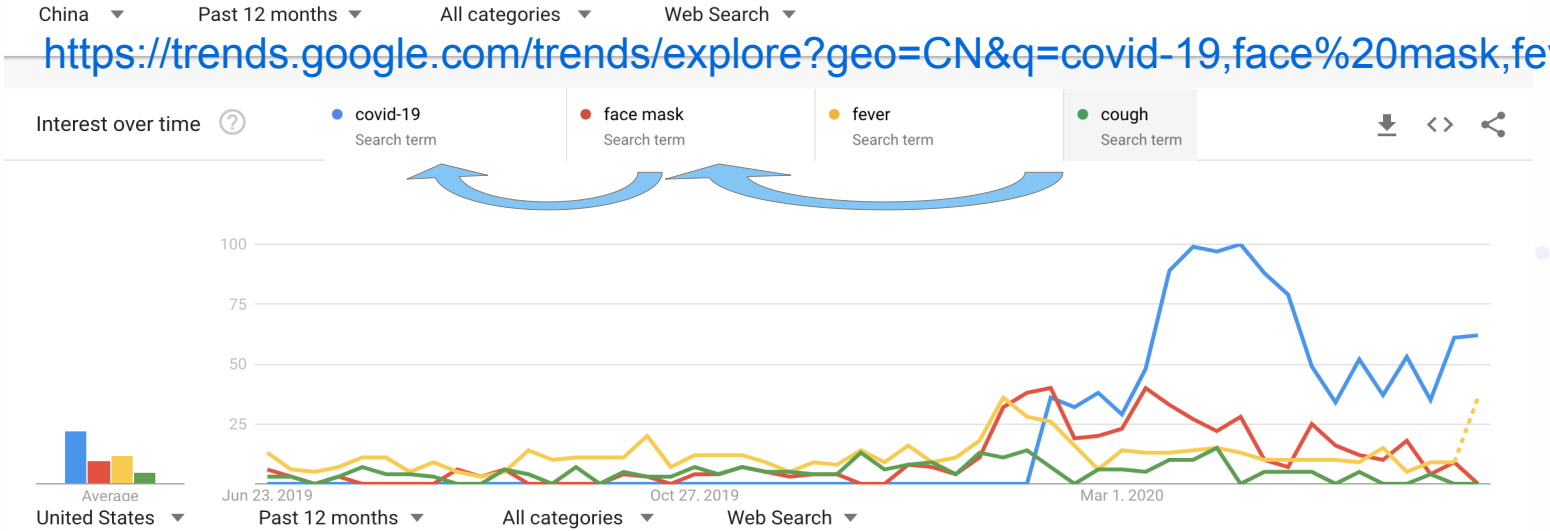



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# What do search queries talk about?

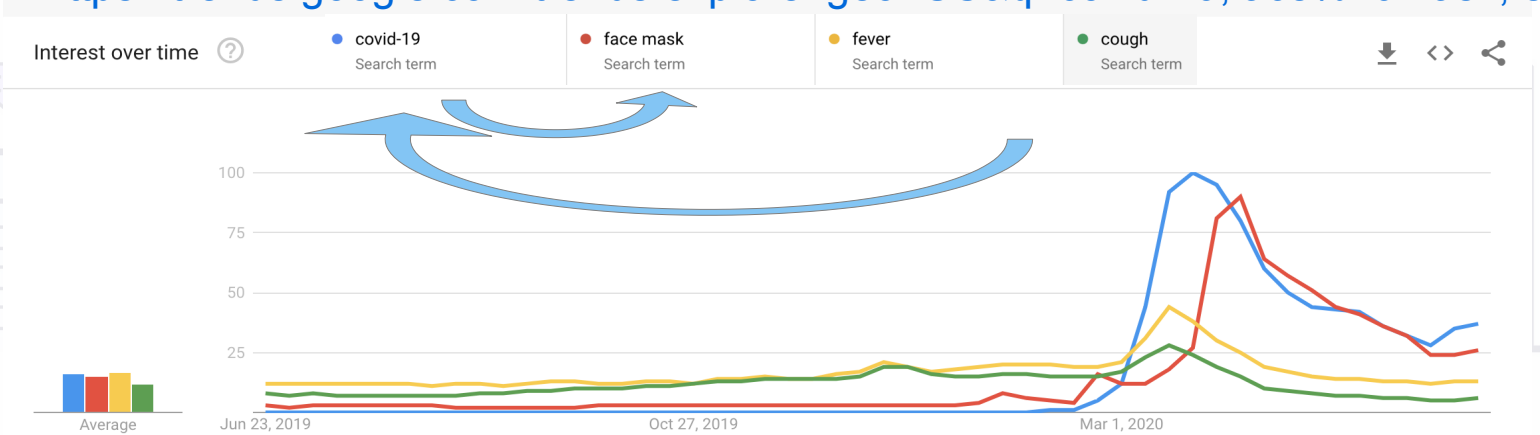
China

https://trends.google.com/trends/explore?geo=CN&q=covid-19,face%20mask,fever,cough




US

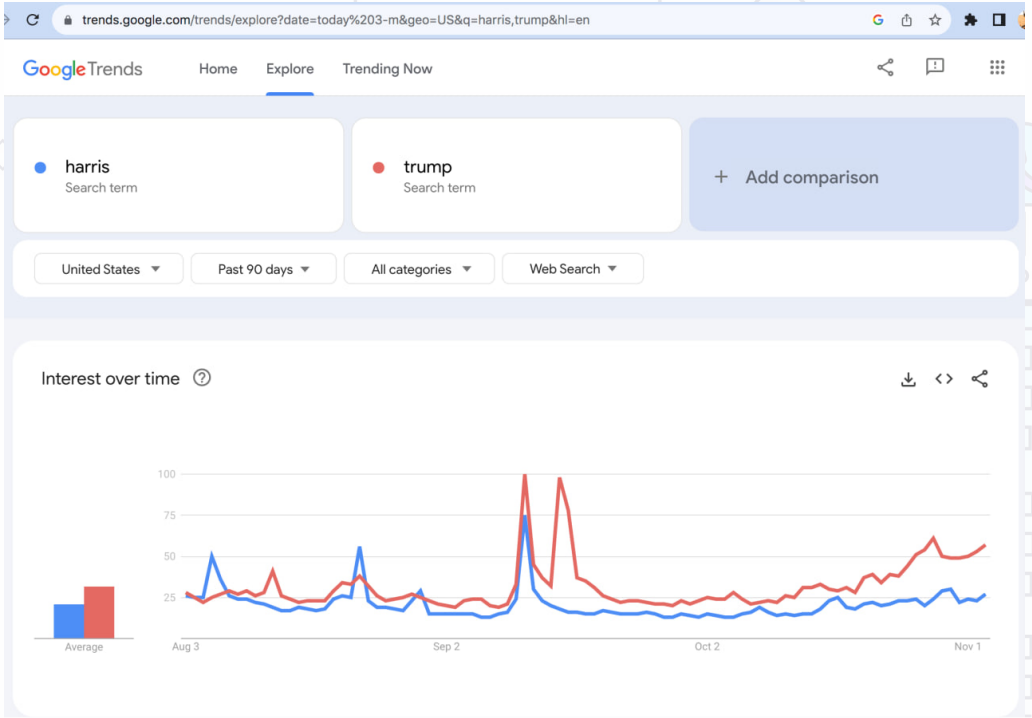
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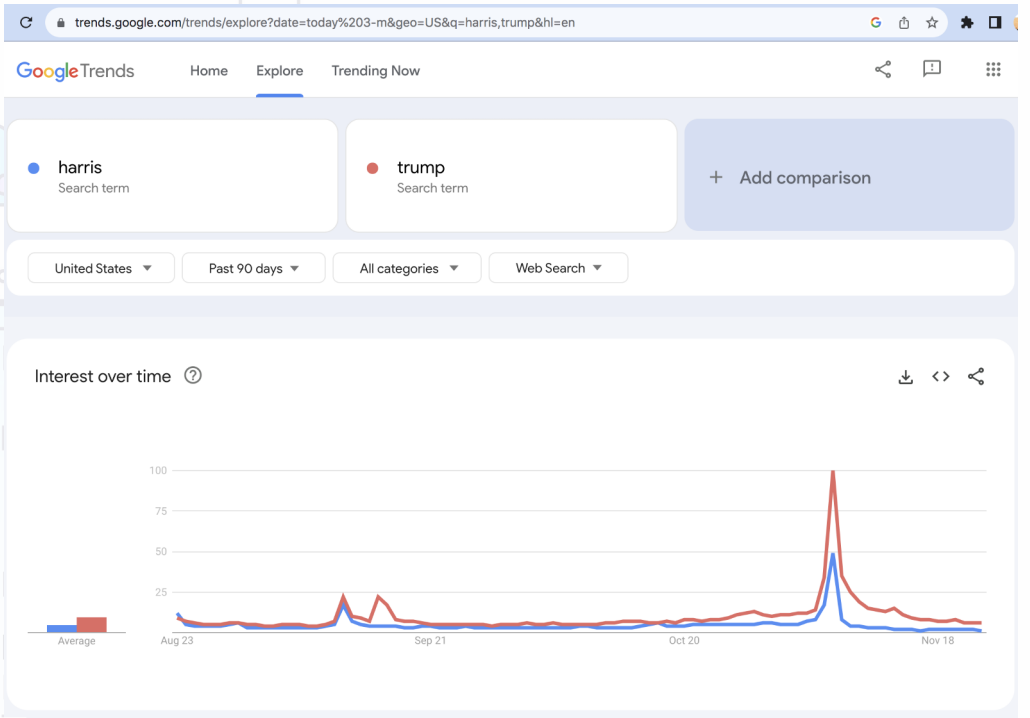

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# What do search queries talk about?

https://trends.google.com/trends/explore?date=today%203-m&geo=US&q=harris,trump&hl=en




Before US elections on November 5, 2024




On the day of elections and after

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# What can be learned about the psyche of a society from its texts?

## Cognitive Distortions (Cognitive-Behavioral Therapy)

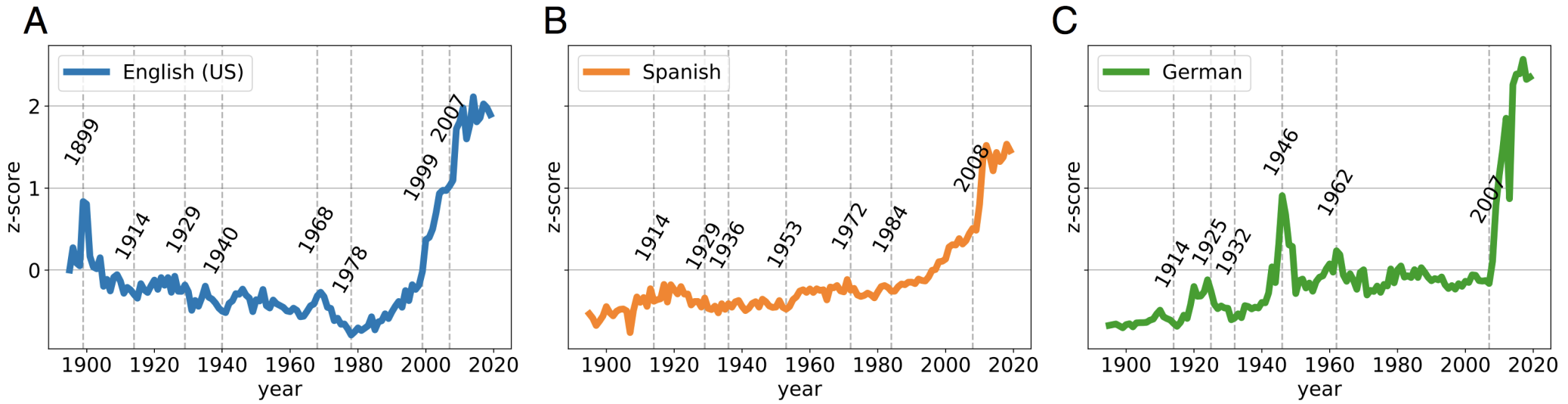
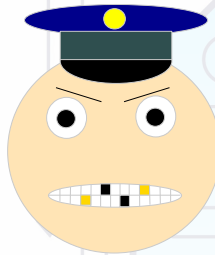



**Fig. 2.** (*A*–*C*) Median z scores of time series of CDS n-gram prevalence from 1855 to 2020 (125 y) in US English (*A*), Spanish (*B*), and German (*C*) with year markers added for major historical events. All time series reveal stable or declining levels for most of the 20th century followed by a sharp surge of cognitive distortions in the past three decades. US English shows declining levels from 1899 to 1978, with minor peaks around 1914 and 1940 (World War I and World War II) and notably 1968. This decline is followed by a surge of CDS prevalence starting in 1978 that continues to 2019. For Spanish we find stable levels from 1895 to the early 1980s at which point a trend occurs toward higher CDS prevalence levels above any of those previously observed. German shows stable CDS prevalence levels, with the exception of strong peaks around and after World War I and World War II, until 2007 at which point a sudden surge occurs.

Source: https://www.pnas.org/doi/pdf/10.1073/pnas.2102061118

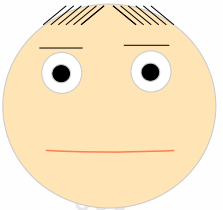
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# Enforcing/preventing manipulation (applied “belief engineering”)

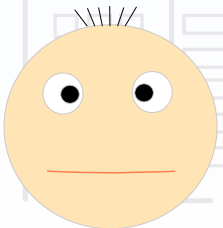


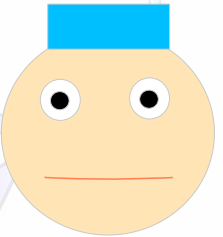
PsyOps,
Social Engineering
Scam





Sales,
Marketing,
Ads





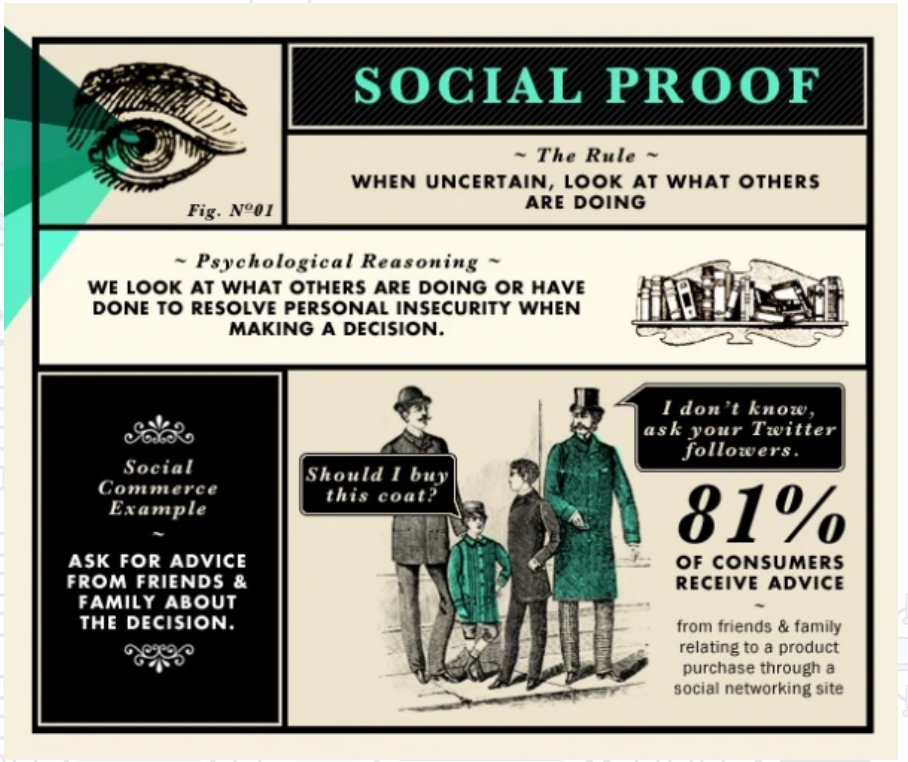
Psychology,
Psychotherapy,
Psychoanalysis



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# Robert Cialdini: Social Proof

Social proof is a psychological and social phenomenon wherein people copy the actions of others in choosing how to behave in a given situation. The term was coined by Robert Cialdini in his 1984 book Influence: Science and Practice, and the concept is also known as informational social influence.



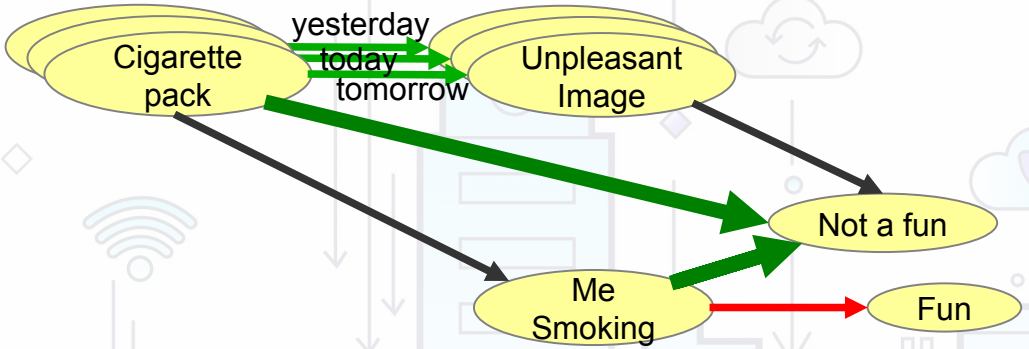

https://www.amazon.com/Influence-Practice-Robert-B-Cialdini/dp/0205609996

*1984*

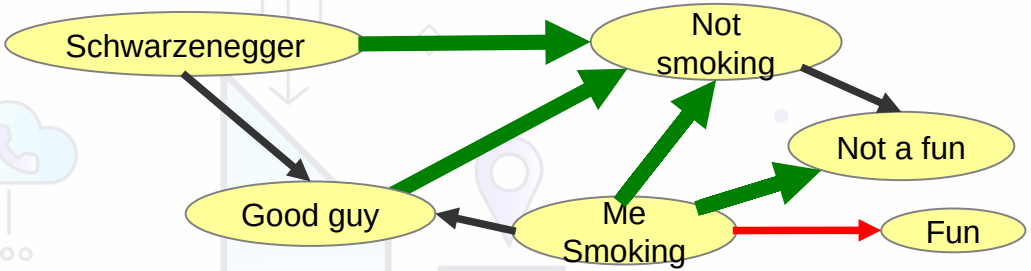
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# Basic technique of applied “belief engineering”

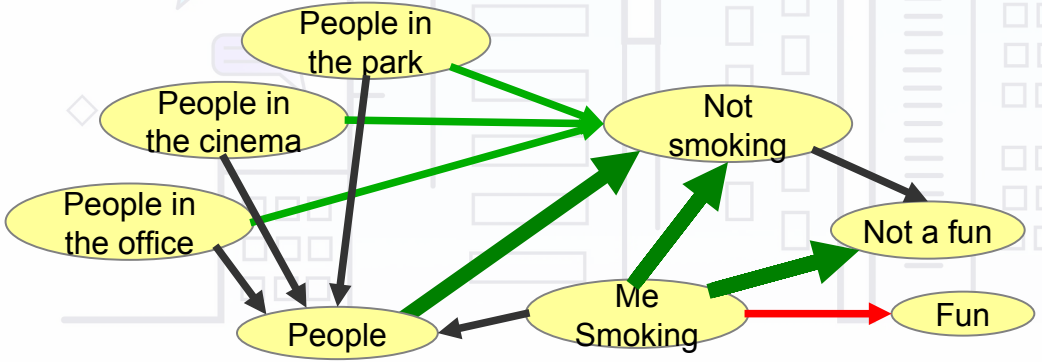
## “Redundant temporal evidence”



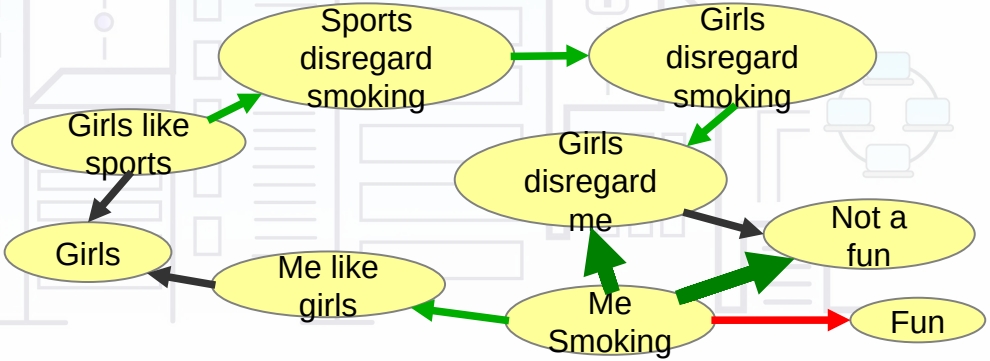

## “Highly valable social evidence”




## “Redundant social evidence”



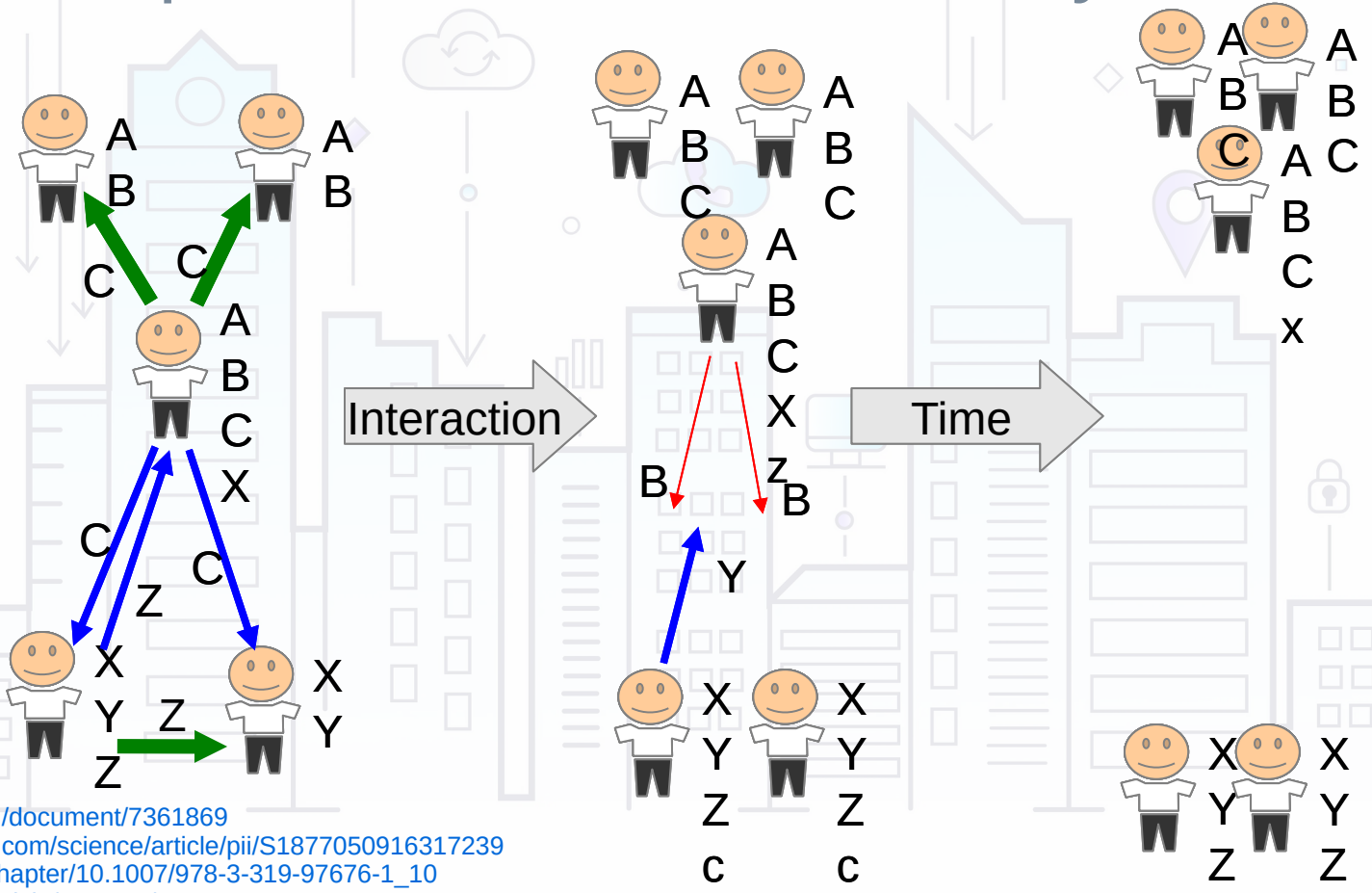

## “Implicit evidence injection”




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# Social evidence-based cognitive-behavioral model

## Helps to understand social dynamics

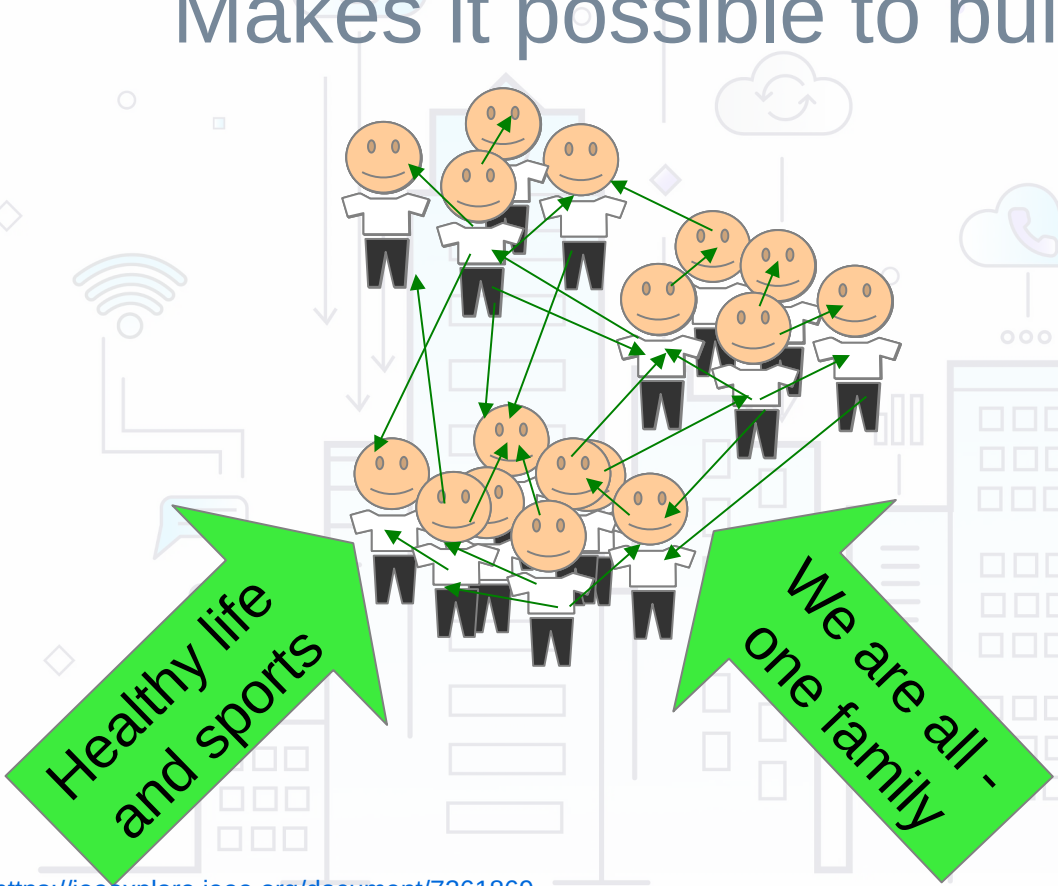


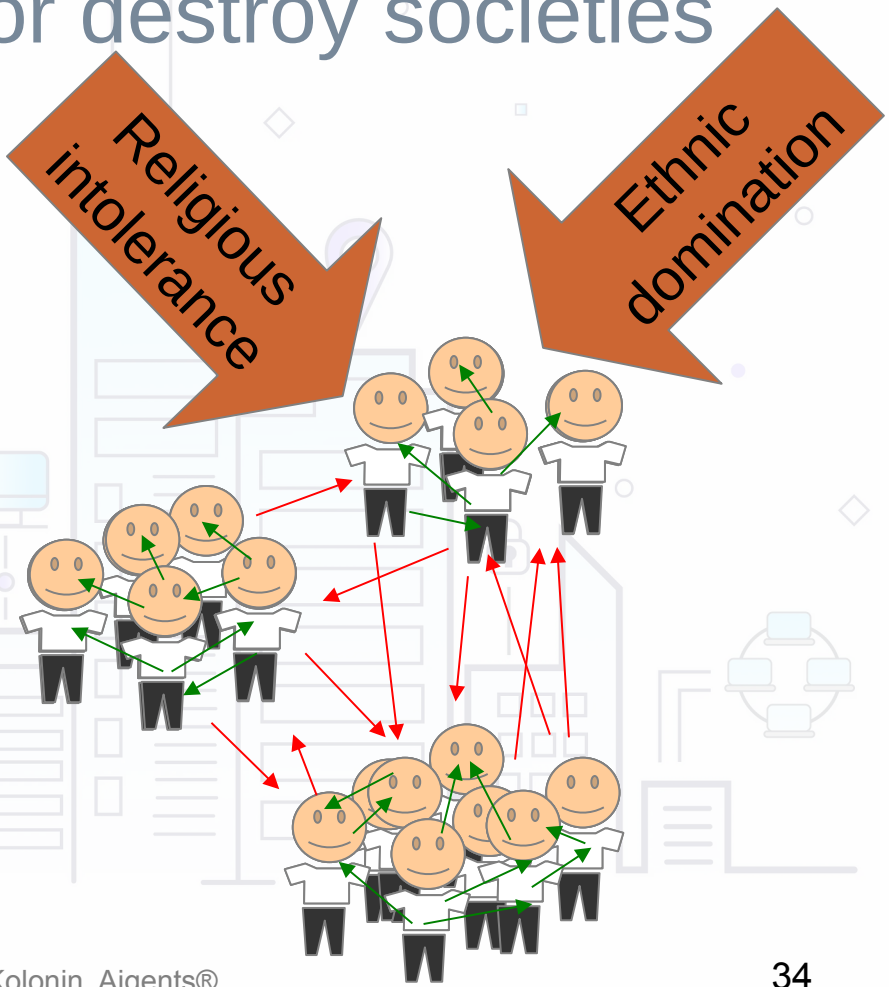

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# Social evidence-based cognitive-behavioral model

## Makes it possible to build or destroy societies

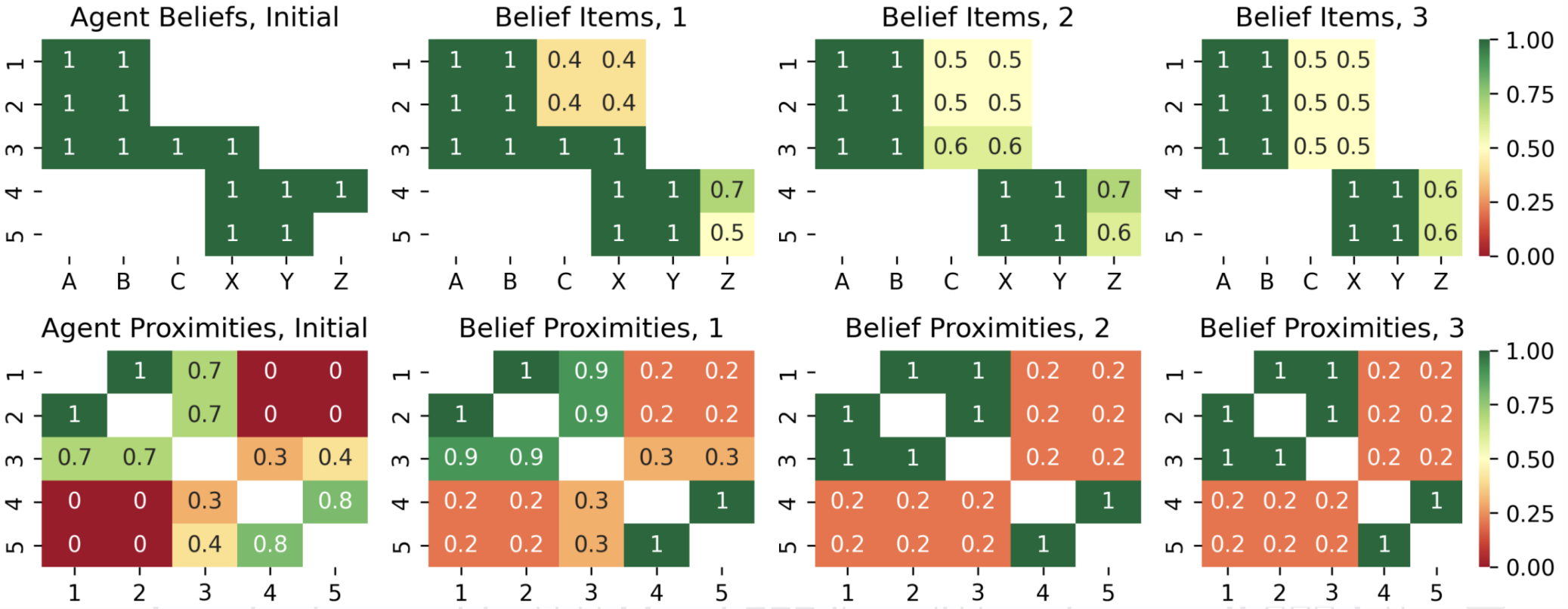



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# Simulation: socially limited, cognitively unlimited

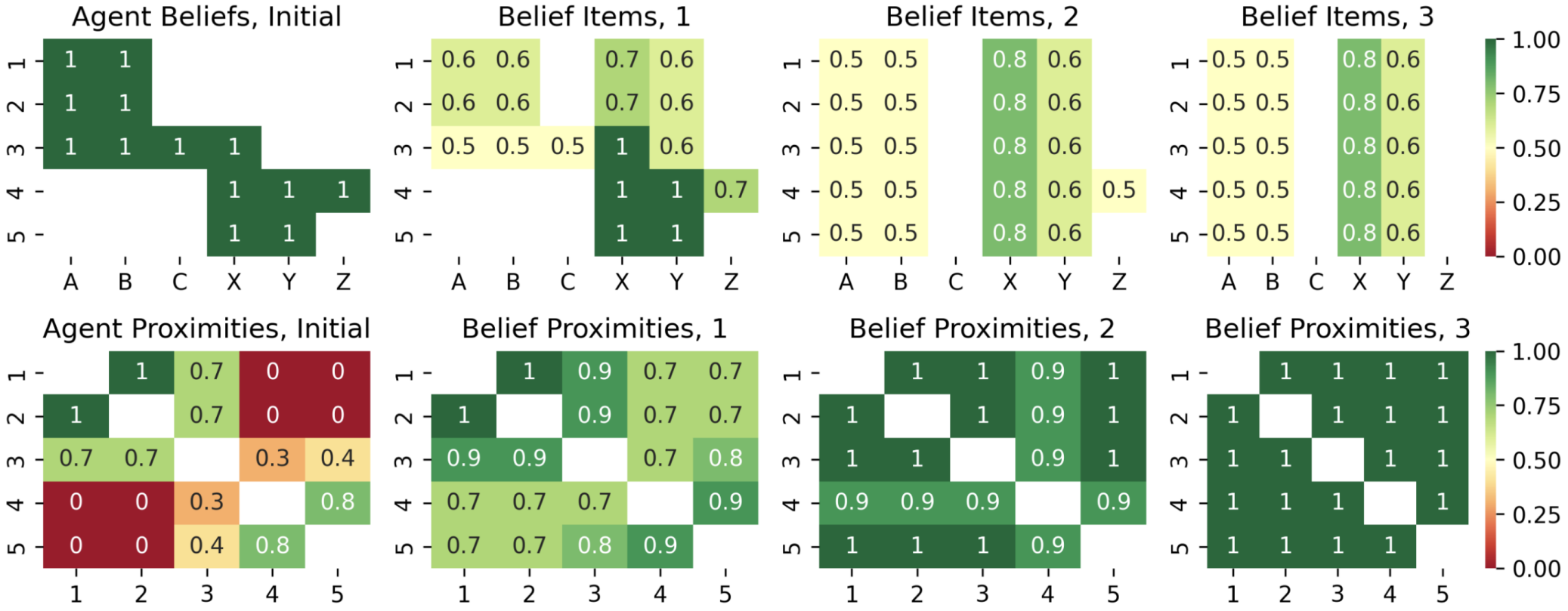



Results of three rounds of multi-agent simulation with the forgetting threshold peer threshold = 0.5, the social relationship threshold forgetting threshold = 0.0 (limiting social connections K without constraint on belief capacity L). Top row: values of belief items A, B, C, X, Y, Z for five agents 1, 2, 3, 4, 5. Bottom row: social proximity matrices between agents, estimated as the cosine similarity of their beliefs. From left to right: the initial state before the simulation, and then the updated states of beliefs and proximities after three subsequent rounds of multi-agent communication.

https://link.springer.com/article/10.1186/s40708-026-00294-1

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# Simulation: socially unlimited, cognitively limited




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